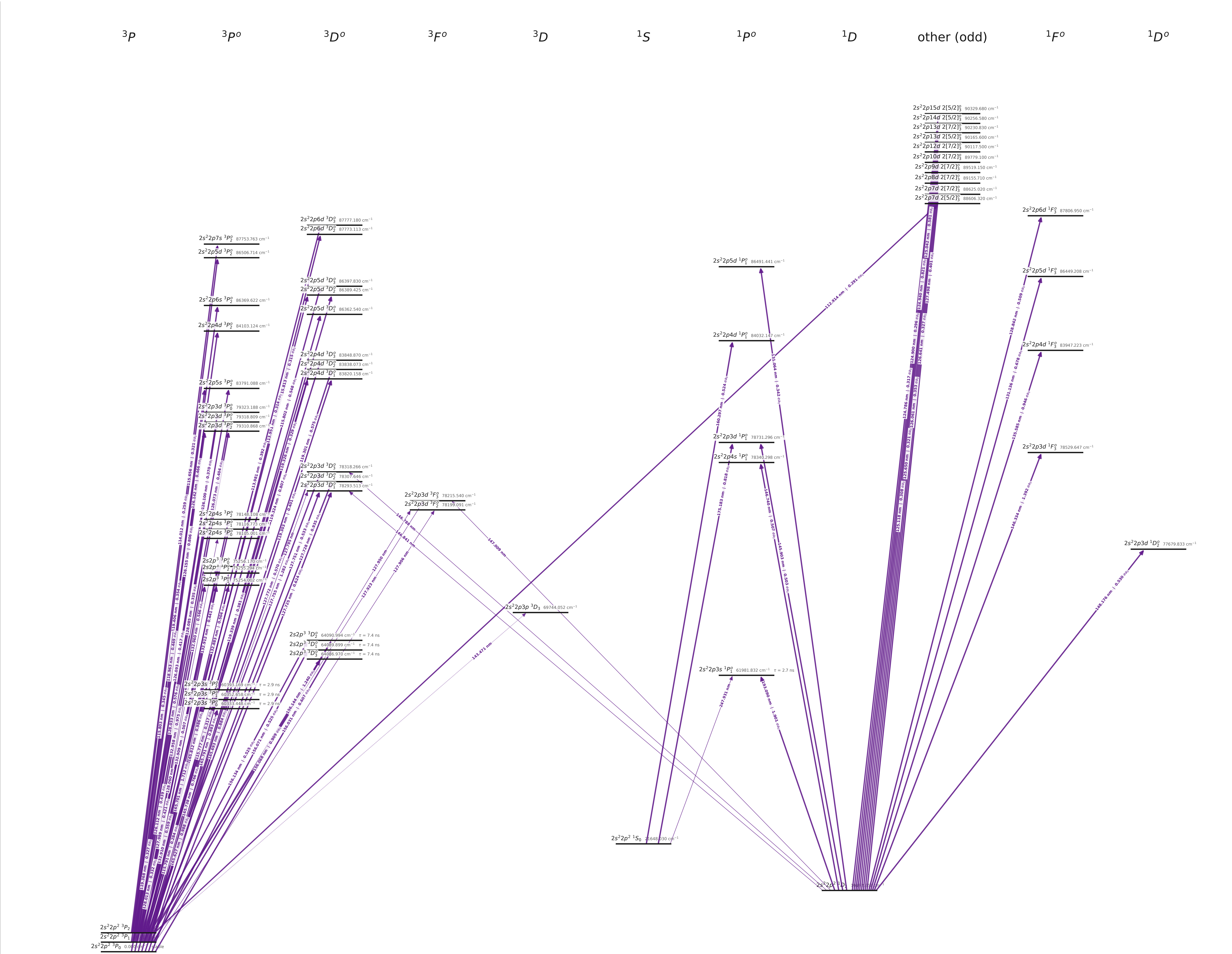


Carbon

C (C I). Strongest classified transitions below 2 μm (NIST): 59 levels, 80 lines.

C+ ionisation limit 90820.35 cm⁻¹ = 11.26029 eV (110.107 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[U er U] (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
2s ² 2p ² 3P ₂ – 2s ² 2p3p 3D ₃	143.4707		2089.5726	–	–	–	–	two-photon excitation of ground-state carbon atoms at 2 x...
2s ² 2p ² 3P ₁ – 2s ² 2p3s 3P ₂ ^o	165.6267	–	1810.0495	0.988	8.700e+07	54.88	25.2	165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...
2s ² 2p ² 3P ₀ – 2s ² 2p3s 3P ₁ ^o	165.6928	–	1809.3272	0.884	1.160e+08	54.88	33.6	165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...
2s ² 2p ² 3P ₂ – 2s ² 2p3s 3P ₂ ^o	165.7008	–	1809.2402	1.712	2.610e+08	54.88	75.7	165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...
2s ² 2p ² 3P ₁ – 2s ² 2p3s 3P ₁ ^o	165.7379	–	1808.8350	0.766	8.700e+07	54.88	25.2	165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...
2s ² 2p ² 3P ₁ – 2s ² 2p3s 3P ₀ ^o	165.7907	–	1808.2591	0.883	3.470e+08	54.88	100	165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...
2s ² 2p ² 3P ₂ – 2s ² 2p3s 3P ₀ ^o	165.8121	–	1808.0257	0.986	1.440e+08	54.88	41.8	165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...
2s ² 2p ² 1D ₂ – 2s ² 2p3s 1P ₁ ^o	193.0905	–	1552.6004	1.901	3.390e+08	58.95	91.5	193.09 nm line: carbon line used in white-dwarf and comet...
2s ² 2p ² 1S ₀ – 2s ² 2p3s 1P ₁ ^o	247.9310	247.8561	1209.1770	–	–	58.95	–	2478.56 Å

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹³C hyperfine structure (I = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2s ² 2p ² 3P ₁	2.838(17)	–	F=1/2: -2.84 F=3/2: +1.42
2s ² 2p ² 3P ₂	149.055(10)	–	F=3/2: -223.58 F=5/2: +149.06

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
2s ² 2p ² 3P ₁ – 2s ² 2p3s 3P ₂ ^o	165.627	13-12	-3028(1049)
2s ² 2p ² 3P ₀ – 2s ² 2p3s 3P ₂ ^o	165.693	13-12	-3418(1049)
2s ² 2p ² 3P ₂ – 2s ² 2p3s 3P ₂ ^o	165.701	13-12	-2908(1049)
2s ² 2p ² 3P ₁ – 2s ² 2p3s 3P ₁ ^o	165.738	13-12	-2578(1049)
2s ² 2p ² 3P ₁ – 2s ² 2p3s 3P ₀ ^o	165.791	13-12	-3058(1049)
2s ² 2p ² 3P ₂ – 2s ² 2p3s 3P ₀ ^o	165.812	13-12	-3118(1049)
2s ² 2p ² 1S ₀ – 2s ² 2p3s 1P ₁ ^o	247.931	13-12	-4677(60)
2s ² 2p ² 1S ₀ – 2s ² 2p3s 1P ₁ ^o	247.931	14-12	-8814(60)

Shift = ν(first isotope) – ν(second isotope).